

Reduced Projected-Cross Moment Factorization

Exact Small-Matrix Closure and a Weak-Mode Conditioning Barrier

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Abstract

RH-94 carries a source packet through the complete frozen prefix by selecting four left singular directions of the ambient projected-cross matrix $K = (I - VV^*)GV$. This paper asks whether that ambient cross SVD can be replaced by packet-scale data.

For an orthonormal rank- r packet V , put $A = V^*GV$, $M_j = V^*G^jV$, and $K = (I - VV^*)GV$. We prove the projected-cross Gram identity

$$K^*K = M_2 - A^2$$

and the cross-cubic identity

$$K^*GK = M_3 - M_2A - AM_2 + A^3.$$

If $K^*K = R\Sigma^2R^*$ on a positive selected subspace, then $Q = KR\Sigma^{-1}$ gives the left cross directions, $V^*GQ = R\Sigma$, and

$$Q^*GQ = \Sigma^{-1}R^*(K^*GK)R\Sigma^{-1}.$$

Hence the complete $(r + k)$ -dimensional Ritz matrix is determined by the first three packet moments, and the ambient cross SVD is algebraically unnecessary. We also state cutoff reconstruction and separated-projector perturbation bounds.

The exact reduction exposes a numerical obstruction. In a 384-bit endpoint audit of 120 source-seeded updates, the fourth/first cross singular ratio falls to 2.77×10^{-11} ; five updates lie below 10^{-8} . Normal equations square this condition number, while the moment identities subtract nearly equal matrices. Raw inverse-singular-value reconstruction loses orthogonality in eight updates, and a binary64 moment-only compression fails a 10^{-3} relative criterion in fifty. Nevertheless, projecting and QR-stabilizing the reconstructed directions, followed by direct small Ritz compression, is tail-stable: all 120 updates agree with the ambient-SVD tail within a factor 1.000000529, and all ten RH-94 endpoint gates remain green.

Thus the ambient cross SVD can be removed, but naive moment-only binary64 closure is rejected. Weak modes must be quotiented or certified by energy, not inverted blindly. No all-level conditioning theorem, Hilbert–Polya operator, zero identification, or Riemann Hypothesis result is claimed.

Keywords: projected-cross Gramian; moment factorization; normal equations; weak singular mode; Ritz method; validated numerics.

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1 Introduction

The source-seeded horizon chain of RH-94 replaces late ambient packet resets by recursive low-dimensional Ritz corrections [6]. At each update, however, it computes the leading left singular directions of

$$K = (I - VV^*)GV,$$

where G is the current positive semidefinite memory Gramian and V is the incoming rank- r packet. Although K has only r columns, its left singular vectors live in the ambient state space.

There are two different reduction questions.

1. Can the singular *selection* be performed in r dimensions?
2. Can the entire corrected Ritz matrix be assembled from packet moments, without an ambient cross SVD?

Both have exact affirmative answers. The right singular data of K are the spectral data of K^*K , and this matrix is the moment difference $V^*G^2V - (V^*GV)^2$. A second identity expresses K^*GK through moments up to order three. Together these formulas give the selected complement directions and every block of the small Ritz compression.

Exact algebra is not the end of the problem. Forming normal equations squares the condition number, reconstructing left vectors divides by selected singular values, and the moment formulas are differences of matrices that can be individually much larger than the cross terms. These familiar numerical hazards [2, 3] become decisive in the archived chain because the fourth cross mode can be almost null.

The resulting paper has a deliberately mixed conclusion. The ambient cross SVD is not structurally necessary: an $r \times r$ factorization plus one cross-column action suffices, and QR stabilization preserves all tail certificates. The stronger claim that binary64 moments alone provide a stable closure is false on the current data. This negative branch is useful: it identifies the precise mode that the next theorem must quotient rather than recover geometrically.

2 Exact projected-cross identities

Let $G = G^* \geq 0$ act on a finite-dimensional Hilbert space $\mathcal{K}$, and let $V : \mathbb{C}^r \rightarrow \mathcal{K}$ be an isometry. Write

$$P = VV^*, \quad A = V^*GV, \quad M_j = V^*G^jV,$$

and define

$$K = (I - P)GV = GV - VA. \tag{1}$$

Theorem 2.1 (Projected-cross Gram identity). *The cross normal matrix is*

$$C := K^*K = M_2 - A^2 \geq 0. \tag{2}$$

Proof. Since $I - P$ is an orthogonal projector,

$$K^*K = V^*G(I - P)GV = V^*G^2V - V^*GVV^*GV = M_2 - A^2.$$

Positivity follows from the first expression. □

Theorem 2.2 (Cross-cubic identity). *The Gramian compressed between cross columns is*

$$N := K^*GK = M_3 - M_2A - AM_2 + A^3. \tag{3}$$

Proof. Using $K = GV - VA$ and $K^* = V^*G - AV^*$,

$$\begin{aligned} K^*GK &= (V^*G - AV^*)G(GV - VA) \\ &= V^*G^3V - V^*G^2VA - AV^*G^2V + AV^*GVA \\ &= M_3 - M_2A - AM_2 + A^3. \end{aligned}$$

□

Neither identity asserts that subtractive evaluation is well conditioned. Indeed, when K is weak, M_2 and A^2 can be nearly equal even though each is accurately computed. The same issue is stronger in (3).

3 Reduced moment factorization theorem

Let the positive selected spectral factor of C be

$$CR = R\Sigma^2, \quad R^*R = I_k, \quad \Sigma = \text{diag}(s_1, \dots, s_k), \quad s_k > 0. \quad (4)$$

Define

$$Q = KR\Sigma^{-1}. \quad (5)$$

Theorem 3.1 (Reduced moment factorization theorem). *The frame Q is an isometry with $Q^*V = 0$. Moreover,*

$$V^*GQ = R\Sigma, \quad (6)$$

$$Q^*GQ = \Sigma^{-1}R^*NR\Sigma^{-1}. \quad (7)$$

Consequently the complete Ritz compression in the orthonormal basis $Z = [V, Q]$ is

$$H = Z^*GZ = \begin{pmatrix} A & R\Sigma \\ \Sigma R^* & \Sigma^{-1}R^*NR\Sigma^{-1} \end{pmatrix}, \quad (8)$$

where C and N are determined by M_1, M_2, M_3 through (2) and (3).

Proof. Because $V^*K = 0$, one has $V^*Q = 0$. Also

$$Q^*Q = \Sigma^{-1}R^*K^*KR\Sigma^{-1} = I_k.$$

Next,

$$V^*GK = V^*G(GV - VA) = M_2 - A^2 = C.$$

Hence

$$V^*GQ = CR\Sigma^{-1} = R\Sigma,$$

which proves (6). Equation (7) follows directly from (5) and $N = K^*GK$. Combining the blocks gives (8). $\square$

Corollary 3.2 (Ambient cross-SVD removal). *The leading k left singular directions of K and the full $(r + k)$ -dimensional Ritz matrix can be obtained from an $r \times r$ spectral factorization, packet moments through order three, and the cross-column reconstruction (5). No singular-value decomposition with an ambient left factor is algebraically required.*

Remark 3.3. The ambient action has not disappeared. Direct formation of K uses GV , and moment formation uses repeated actions of G on V . What disappears is the ambient *spectral solve*.

4 Cutoff stability

The inverse Σ^{-1} identifies the conditioning immediately. For a fixed selected right frame R and fixed positive Σ , replacing K by $K + E$ changes the reconstructed frame by

$$\hat{Q} - Q = ER\Sigma^{-1}.$$

Proposition 4.1 (Reconstruction amplification). *If $s_k \geq \tau > 0$, then*

$$\|\hat{Q} - Q\|_F \leq \frac{\|E\|_F}{\tau}. \quad (9)$$

Proof. Use $\|R\Sigma^{-1}\|_2 \leq \tau^{-1}$ and submultiplicativity. $\square$

The right spectral frame also changes if the cross Gram is perturbed. Let C have a squared-singular-value gap

$$\gamma = s_k(K)^2 - s_{k+1}(K)^2 > 0.$$

For a Hermitian perturbation F , the Davis–Kahan theorem controls the selected spectral projectors [1, 5].

Proposition 4.2 (Separated-cutoff projector bound). *If $2\|F\|_2 < \gamma$, then, after matching the leading rank- k spectral clusters,*

$$\left\| \widehat{R}\widehat{R}^* - RR^* \right\|_2 \leq \frac{2\|F\|_2}{\gamma}. \quad (10)$$

The two estimates expose distinct requirements: the cutoff squared gap controls the right frame, while the cutoff singular value controls left-frame reconstruction. A gap can be adequate even when s_k itself is tiny.

4.1 Why normal equations are dangerous here

Suppose $s_1(K)/s_k(K) = \kappa$. The positive spectrum of $C = K^*K$ has condition number κ^2 . At binary64 precision, a ratio $s_k/s_1 \approx 10^{-10}$ places the smallest squared mode far below ordinary relative resolution. Even if a small-matrix SVD repairs a tiny negative eigenvalue of a computed C , reconstructing Q still amplifies cross-column error by 10^{10} .

Moment evaluation adds cancellation indices

$$\chi_2 = \frac{\|M_2\|_F + \|A^2\|_F}{\|C\|_F}, \quad (11)$$

$$\chi_3 = \frac{\|M_3\|_F + \|M_2A\|_F + \|AM_2\|_F + \|A^3\|_F}{\|N\|_F}. \quad (12)$$

Large χ_j do not invalidate the identities; they show why direct binary64 subtraction may not realize them accurately.

5 Validated audit

5.1 Protocol

We replay the 120 width-four source-seeded updates of RH-94 [6]. At each update we form K but replace its ambient left SVD by:

1. an SVD of the $r \times r$ matrix K^*K as a positive-semidefinite repair;
2. the reconstruction $Q = KR\Sigma^{-1}$;
3. projection away from V and a thin QR factorization;
4. direct assembly of the stabilized $(r + 4)$ Ritz compression.

The direct small compression is used for the primary stabilized chain. In parallel, the moment-only matrix (8) is assembled in binary64 and compared with its direct counterpart. The ambient cross-SVD construction is retained only as a diagnostic comparator.

Tail quadratic forms and endpoint ratios are evaluated with exact binary inputs in Arb at 384-bit precision [4]. The positive route gate requires every stabilized reduced tail to lie within a factor 1.0001 of the ambient-SVD tail and every channel endpoint to remain below the RH-94 ratio 1.01. The negative diagnostics mark

$$s_4/s_1 < 10^{-8}, \quad \|Q^*Q - I\|_2 > 10^{-6}, \quad \frac{\|H_{\text{moment}} - H_{\text{direct}}\|_F}{\|H_{\text{direct}}\|_F} > 10^{-3}.$$

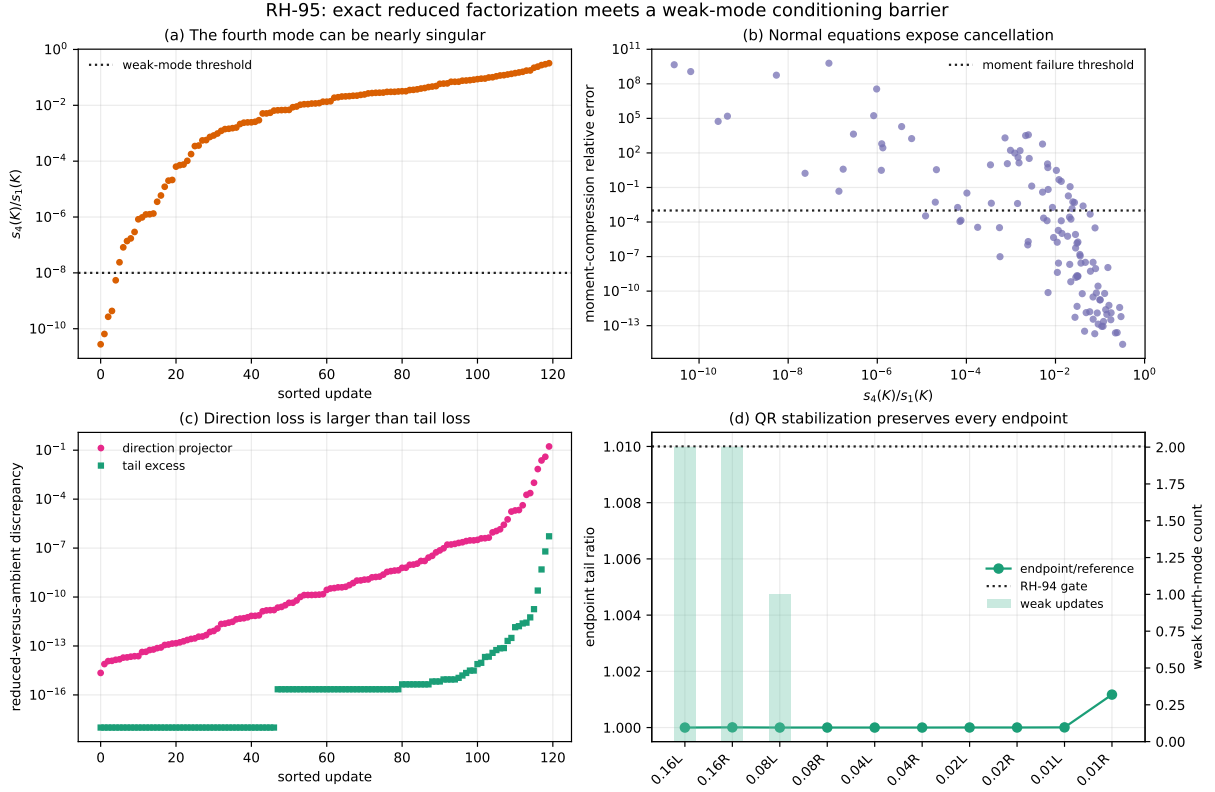


Figure 1: The reduced factorization has two faces. Weak fourth modes make normal equations and moment subtraction ill conditioned, yet QR-stabilized reconstruction preserves all Ritz tail and endpoint gates.

5.2 Positive result: tail-stable QR reconstruction

All 120 stabilized updates are directly Ritz-monotone. The largest reduced/ambient-SVD tail ratio is

$$1.0000005281,$$

and all ten endpoints remain green. The worst endpoint/reference ratio is 1.001172049, essentially unchanged from RH-94. The stabilized packet orthogonality defect is below 2.77×10^{-15} , and the compressed dimension remains at most 11.

The selected direction projectors need not be equally close. Their largest operator-norm distance is 0.1693, whereas the largest corrected packet projector distance is 1.41×10^{-4} and the tail discrepancy is much smaller still. Weak directions are therefore geometrically unstable but energetically almost invisible at these updates.

5.3 Negative result: moment-only binary64 closure

Five updates have $s_4(K)/s_1(K) < 10^{-8}$, and the minimum ratio is

$$2.7686 \times 10^{-11}.$$

Raw reconstruction exceeds the orthogonality-failure threshold in eight updates, reaching a defect 0.9581. QR repairs the frame, but it cannot make the moment subtraction itself accurate.

The binary64 moment compression fails its 10^{-3} relative criterion in fifty of the 120 updates. The largest compressed-moment relative error is 6.17×10^9 , while the cross-cubic relative discrepancy can reach 1.97×10^{13} . These values reject a naive implementation of (8); they do not contradict the exact identities.

6 Interpretation: a weak-mode quotient

The data suggest that demanding a stable geometric fourth direction is the wrong next goal. In the weakest updates, many directions can represent the almost-null cross mode, but the corrected Ritz tail barely changes. The relevant equivalence relation should therefore be energetic:

$Q \sim \tilde{Q}$ if their corrected rank- r Ritz tails differ by at most a certified tolerance.

A future weak-mode quotient theorem should bound the loss caused by discarding or arbitrarily completing modes below a cutoff. Such a bound will need more than discarded cross energy alone. It should include:

- the unselected cross Frobenius energy;
- a gap or residual for the leading rank- r compressed Ritz cluster;
- the complement diagonal energy available in omitted directions;
- the current tail scale, so the statement is relative to the target.

If this bound is small, the chain may use adaptive width and avoid inversion of numerically meaningless singular modes.

7 Claim boundary

The exact projected-cross Gram identity and reduced moment factorization theorem are finite-dimensional algebraic results. The audit additionally supports QR-stabilized tail equivalence on ten frozen channels. It does not establish:

1. stable binary64 moment-only closure—the archived experiment rejects that claim;
2. a uniform positive lower bound for the fourth cross singular value or its squared gap;
3. an analytic weak-mode quotient or adaptive-width theorem;
4. removal of the ambient actions needed to form GV , G^2V , or G^3V ;
5. repeated all-level block contraction, the normalization/observability bridge, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

8 Conclusion

The projected-cross step has an exact small-matrix closure. Its right singular data are carried by $M_2 - A^2$, its complement Gram block is carried by moments through order three, and the ambient cross SVD can be replaced by an $r \times r$ factorization plus reconstruction. With projection and QR stabilization, this replacement preserves every audited tail and endpoint certificate.

The same calculation reveals why a stronger moment-only claim would be premature. The fourth cross mode can be ten orders of magnitude below the first, so normal equations, inverse-singular reconstruction, and subtractive moment formulas encounter severe conditioning. The next layer should not try to identify this weak direction more accurately. It should prove that weak cross modes can be quotiented by a certified tail-loss bound.

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